

Mini-Shell Test Suite

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1 Directory Navigation Tests

1.1 Test 1: ls, cd, pwd basic functionality

Description: Verify that basic built-in commands (ls, cd, pwd) work properly inside the mini-shell.

```
hobbbbit31@Smaug:~/Desktop/mini-shell$ ./mini-shell
/home/hobbbbit31/Desktop/mini-shell$ ls
include  Makefile  mini-shell  mini-shell.c  techincal_report.pdf  technical_report.md  user_documentation.md  user_documentation.pdf
/home/hobbbbit31/Desktop/mini-shell$ cd tests
cd: No such file or directory
/home/hobbbbit31/Desktop/mini-shell$ cd
/home/hobbbbit31$ cd Desktop
/home/hobbbbit31/Desktop$ cd mini-shell
/home/hobbbbit31/Desktop/mini-shell$
```

Figure 1: Testing ls and cd commands

2 Redirection Tests

2.1 Test 2: Output Redirection

Description: Check whether command > file correctly writes output to a file.

```
/home/hobbbbit31/Desktop/mini-shell$ ls > out.txt
/home/hobbbbit31/Desktop/mini-shell$ cat<out.txt
include
Makefile
mini-shell
mini-shell.c
out.txt
techincal_report.pdf
technical_report.md
user_documentation.md
user_documentation.pdf
/home/hobbbbit31/Desktop/mini-shell$
```

Figure 2: Testing output redirection using ;

2.2 Test 3: Input Redirection

Description: Ensure that command < file works as expected.

```
/home/hobbbbit31/Desktop/mini-shell$ echo hello >> out.txt
/home/hobbbbit31/Desktop/mini-shell$ cat < out.txt
include
Makefile
mini-shell
mini-shell.c
out.txt
techincal_report.pdf
technical_report.md
user_documentation.md
user_documentation.pdf
hello
/home/hobbbbit31/Desktop/mini-shell$
```

Figure 3: Testing input redirection using ;

3 Navigation and Built-in Commands

3.1 Test 4: cd with full path and relative path

```
/home/hobbbbit31/Desktop/mini-shell $ pwd
/home/hobbbbit31/Desktop/mini-shell
/home/hobbbbit31/Desktop/mini-shell $ cd ..
/home/hobbbbit31/Desktop $ cd ..
/home/hobbbbit31 $ cd ..
/home $ cd ..
/ $ cd home
/home $ cd hobbbbit31
/home/hobbbbit31 $ cd Desktop
/home/hobbbbit31/Desktop $ cd mini-shell
/home/hobbbbit31/Desktop/mini-shell $
```

Figure 4: Testing cd with relative and absolute paths

4 Foreground and Background Jobs

4.1 Test 5: sleep foreground test

Description: Verify that the shell waits for the foreground process to finish.

```
/home/hobbbbit31/Desktop/mini-shell $ sleep 10
█
```

Figure 5: Testing sleep in foreground

5 Tokenizer Tests

5.1 Test 6: Quoted Strings

Description: Check that quoted arguments are parsed correctly and unmatched quotes produce an error.

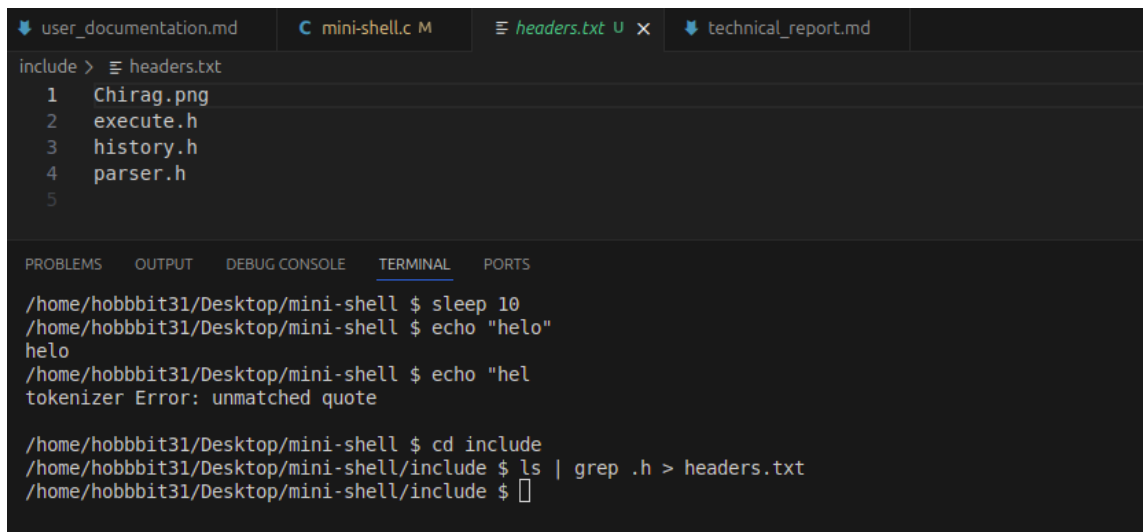
```
/home/hobbbbit31/Desktop/mini-shell $ sleep 10
/home/hobbbbit31/Desktop/mini-shell $ echo "helo"
helo
/home/hobbbbit31/Desktop/mini-shell $ echo "hel
tokenizer Error: unmatched quote

/home/hobbbbit31/Desktop/mini-shell $
```

Figure 6: Tokenizer handling quoted and unmatched quoted strings

6 File Filtering Tests

6.1 Test 7: grep with redirection



The screenshot shows a code editor interface. At the top, there are tabs for 'user_documentation.md', 'mini-shell.c M', 'headers.txt U x', and 'technical_report.md'. Below the tabs, the 'include' directory is open, showing a list of files: '1 Chirag.png', '2 execute.h', '3 history.h', '4 parser.h', and '5'. At the bottom, the 'TERMINAL' tab is active, displaying the following commands and output:

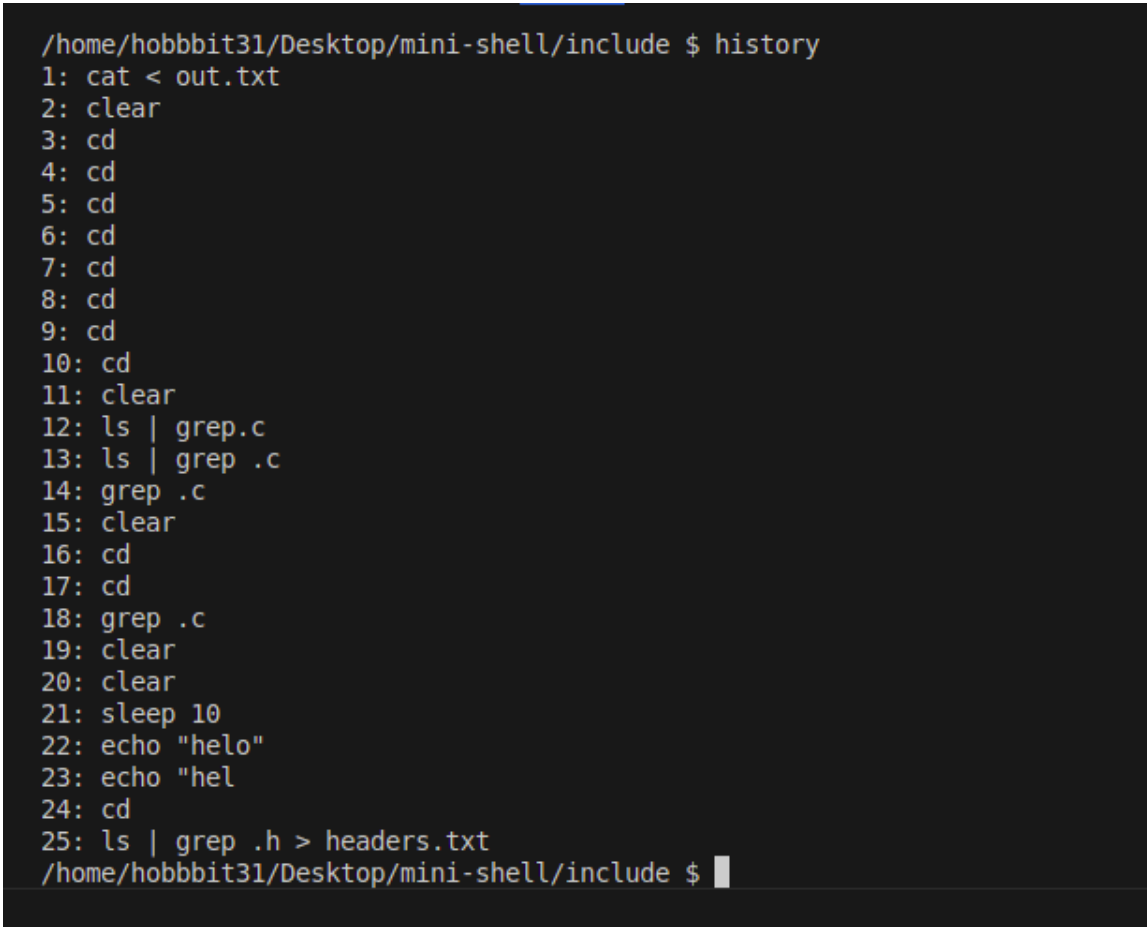
```
/home/hobbbbit31/Desktop/mini-shell $ sleep 10
/home/hobbbbit31/Desktop/mini-shell $ echo "helo"
helo
/home/hobbbbit31/Desktop/mini-shell $ echo "hel
tokenizer Error: unmatched quote

/home/hobbbbit31/Desktop/mini-shell $ cd include
/home/hobbbbit31/Desktop/mini-shell/include $ ls | grep .h > headers.txt
/home/hobbbbit31/Desktop/mini-shell/include $
```

Figure 7: Testing grep and redirection to headers.txt

7 History Command Tests

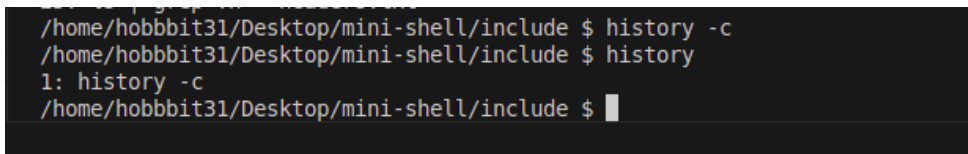
7.1 Test 8: history list functionality

A terminal window with a dark background and light-colored text. The prompt is `/home/hobbbbit31/Desktop/mini-shell/include $`. The command `history` has been entered, and the output is a list of 25 commands, each preceded by a number from 1 to 25. The commands include `cat < out.txt`, `clear`, `cd`, `ls | grep .c`, `grep .c`, `sleep 10`, `echo "helo"`, `echo "hel`, and `ls | grep .h > headers.txt`. The prompt is followed by a cursor.

```
/home/hobbbbit31/Desktop/mini-shell/include $ history
1: cat < out.txt
2: clear
3: cd
4: cd
5: cd
6: cd
7: cd
8: cd
9: cd
10: cd
11: clear
12: ls | grep .c
13: ls | grep .c
14: grep .c
15: clear
16: cd
17: cd
18: grep .c
19: clear
20: clear
21: sleep 10
22: echo "helo"
23: echo "hel
24: cd
25: ls | grep .h > headers.txt
/home/hobbbbit31/Desktop/mini-shell/include $
```

Figure 8: Testing history command

7.2 Test 9: history -c

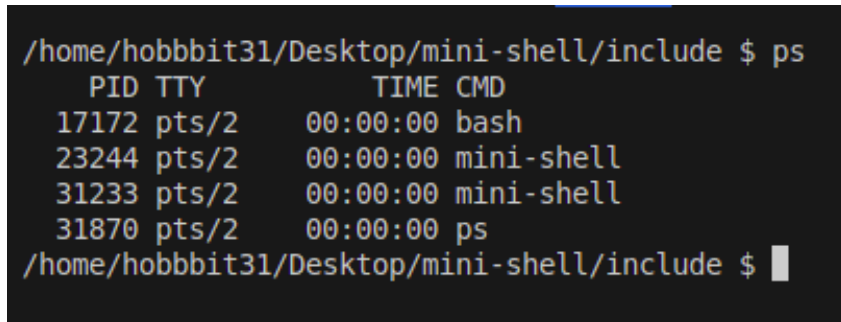
A terminal window with a dark background and light-colored text. The prompt is `/home/hobbbbit31/Desktop/mini-shell/include $`. The command `history -c` has been entered. The prompt is followed by a cursor.

```
/home/hobbbbit31/Desktop/mini-shell/include $ history -c
/home/hobbbbit31/Desktop/mini-shell/include $ history
1: history -c
/home/hobbbbit31/Desktop/mini-shell/include $
```

Figure 9: Testing history -c command

8 Process Listing Test

8.1 Test 10: ps command



A terminal window showing the execution of the 'ps' command. The prompt is '/home/hobbbbit31/Desktop/mini-shell/include \$'. The output is a table with four columns: PID, TTY, TIME, and CMD. The table lists four processes: bash (PID 17172), mini-shell (PID 23244), mini-shell (PID 31233), and ps (PID 31870). All TTYs are 'pts/2' and all times are '00:00:00'. The prompt after the command is '/home/hobbbbit31/Desktop/mini-shell/include \$' followed by a cursor.

PID	TTY	TIME	CMD
17172	pts/2	00:00:00	bash
23244	pts/2	00:00:00	mini-shell
31233	pts/2	00:00:00	mini-shell
31870	pts/2	00:00:00	ps

Figure 10: Testing ps inside mini-shell